Summary

This document is the Interim report of the project “Dr. Jigree” and highlights the progress that is made up until the date of the submission of the Interim Report. This report aims to show the understanding of the system in comparison to when it was previously proposed. This report successfully shows the research done and the progress made in terms of the development, which indicates the level of understanding developed during the process. The comparison with similar products is clearly tabled with differences. In addition to the research done on the topic, the development progress has also been clearly tracked and analyzed where different diagrams have been used to demonstrate the system and have a clearer understanding. The development methodology is also highlighted with the reasoning of picking the module. The interim report then ended with the future work which highlighted how the system would further need to be developed.

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1. Introduction

1.1. Problem Scenario

With rapid urbanization in recent decades, sedentary work schedules, along with the rise of fast foods, there have been significant changes in people's daily lifestyles. Different fatal diseases like obesity, diabetes, cardiovascular diseases, hypertension, and stress disorders continue to rise, with many conditions developing due to lack of early attention or poor health information available. According to the World Health Organization (WHO, 2024), over 1.9 billion adults are overweight and more than 650 million are obese worldwide.

Most of the employees who work regular 9-5 or longer often don't have easy access to doctors and checkups as there is an overlap between the doctor's hours and work. This makes a significant difference in people's access to any health resources or consultation. A CDC report (2023) indicates that 76% of adults do not meet recommended physical activity levels, and nearly 60% fail to monitor essential indicators such as BMI, blood pressure, diet quality, and sleep habits. Even with access to healthcare it can be costly to do regular checkups which leads to people hesitating to reach out to health services fearing the expense tied to the treatment.

Currently available digital tools tend to be fragmented as most of them focus on fitness tracking, meal planning, or schedule tracking, and most usually require a premium subscription. Platforms like MyFitnessPal, Fitbit and Noom offer valuable information but often restrict essential features behind paywalls. As a result, there is a clear gap for an accessible, unified, AI-assisted health management portfolio tool that can provide a personalized health and fitness recommendation tool.

1.2. The Project as a Solution

This project proposes the development of a Health Portfolio Management Application, a centralized digital platform that allows users to log into key health indicators and receive meaningful personalized recommendations. Dr. Jigree will combine basic health assessments like BMI, blood pressure category, lifestyle habits, etc. with an AI-enhanced module capable of generating disease warning, personalized insights on fitness, dietary suggestions as per user's goal along with daily workout routines. The system will integrate both rule-based logic and machine learning models trained on publicly available datasets like NHANES to estimate potential health risks and classify users into general wellness segments. Dr. Jigree offers users a structured health profile, daily action plan, and exportable health summary.

1.3. Aims and Objectives

1.3.1 Main Aim

To develop an intelligent, user-friendly health portfolio management system that takes basic inputs from a user and assesses their lifestyle indicators and provides personalized recommendations for diet, exercise, and overall wellness improvement. Dr. Jigree will also identify potential health risks and warn users to get a professional checkup in that matter if necessary.

1.3.2 Objectives

1. To design and implement a secure user profile module so that individuals can sign up and system collects and store essential health and lifestyle data.

2. To build a health scoring model combining BMI calculations, rule-based metrics, and ML-based risk estimation to classify user health levels.
3. To develop a personalized dietary recommendation engine that generates calorie goals, macronutrient distributions, and daily meal suggestions based on the goal that is selected from users.
4. To integrate an AI-supported workout plan generator tailored to user goals (weight loss, muscle gain, etc.).
5. To provide a dashboard interface summarizing user status, recommended actions and activities, and daily plans.
6. To allow a PDF export functionality where users can download their data and report in pdf.

1.4 Report Structure

1.4.1 Background

The background section consists of information on the usage of the application by the end user and their benefits from the system. It also contains the background of the application itself and how it aims to solve problems faced by the veterinarian sector in Nepal. The literature review section contains reports and features of existing systems on a global scale that are like this project. This section includes screenshots of the features that are to be implemented in the system and compares them with other systems as well.

1.4.2 Development

This section of the report outlines the progress made in the project so far, including the completion of the initial planning phase, the design of the system architecture, and the setup for development. Screenshots of the database schema and system diagrams

have been included to illustrate the progress visually. The development of PawsPulse has been planned according to the sprint schedule, ensuring steady progress toward the project goals.

1.4.3 Analysis of progress

The project has completed its initial planning and requirements gathering phase. The architectural design and database schema have been established, laying a strong foundation for the development phase. While progress remains largely on schedule, particular attention is being given to ensuring robust data validation and security measures are in place, given the sensitive nature of personal details. The implementation of core features, exporting of the report, risk score, and details is proceeding according to the sprint schedule, with regular testing and refinement occurring at each stage.

1.4.4 Future Work

This section outlines the remaining progress needed to further enhance the system, including the implementation of advanced features such as an Emergency Assistance System and predictive analytics tools. Thes plans are in place to refine the user interface based on feedback and conduct comprehensive testing to ensure reliability and security. The team will follow the Agile methodology to manage remaining tasks and ensure timely delivery. Long-term goals involve scaling the system for broader adoption and incorporating emerging technologies to enhance functionality further.

2. Background and Literature Review

2.1 About the End User / Client

The primary users of the system are working adults and students who have limited time to manage their health proactively. These users typically possess basic digital literacy but lack advanced medical knowledge. The system is designed for self-monitoring and general wellness guidance rather than clinical diagnosis.

2.2 Understanding the Project

The project integrates frontend development, backend services, and machine learning components. Users input basic data such as height, weight, activity level, and lifestyle habits. The system processes this data to generate health scores and recommendations using a hybrid AI approach. The system will output the meal plans along with the workout routine to end users.

2.3 Review of Similar Projects

2.3.1 Project 1: MyFitnessPal

MyFitnessPal is a widely used health and fitness tracking application focusing primarily on calorie counting and dietary logging. The strength and limitation of the MyFitnessPal is listed below.

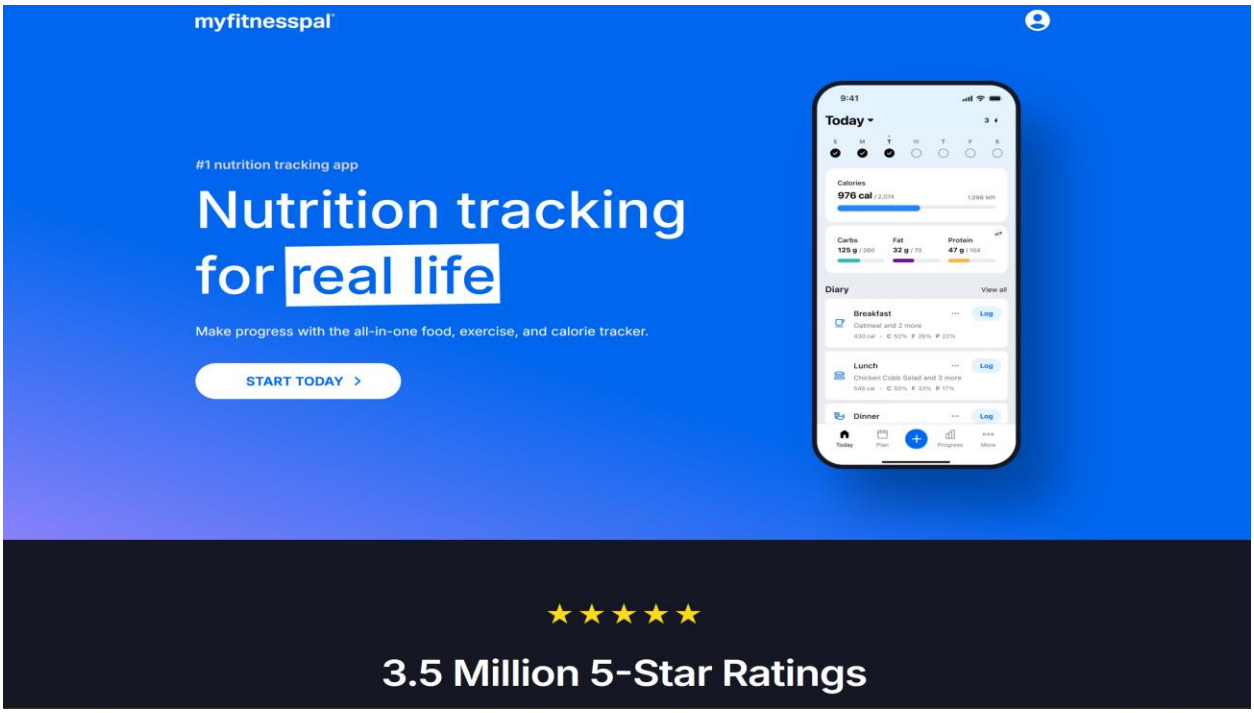


Figure 1: MyFitness Pal

Strengths:

My fitness pal has an extremely large food database with a variety of data along with great tracking features that allow users to track calories intake along with burns. It also has a great user interface catering to a large volume of users.

Limitations:

Although fitness pal comes with extra ordinary features, it has few gaps. All the advanced insights are under a paid subscription. It also doesn't provide AI-driven recommendations, mainly focusing on guidance more than tracking.

2.3.2 Project 2: Fitbit Health Platform

Fitbit provides fitness tracking through wearable devices, focusing on physical activity, sleep, and heart rate monitoring.

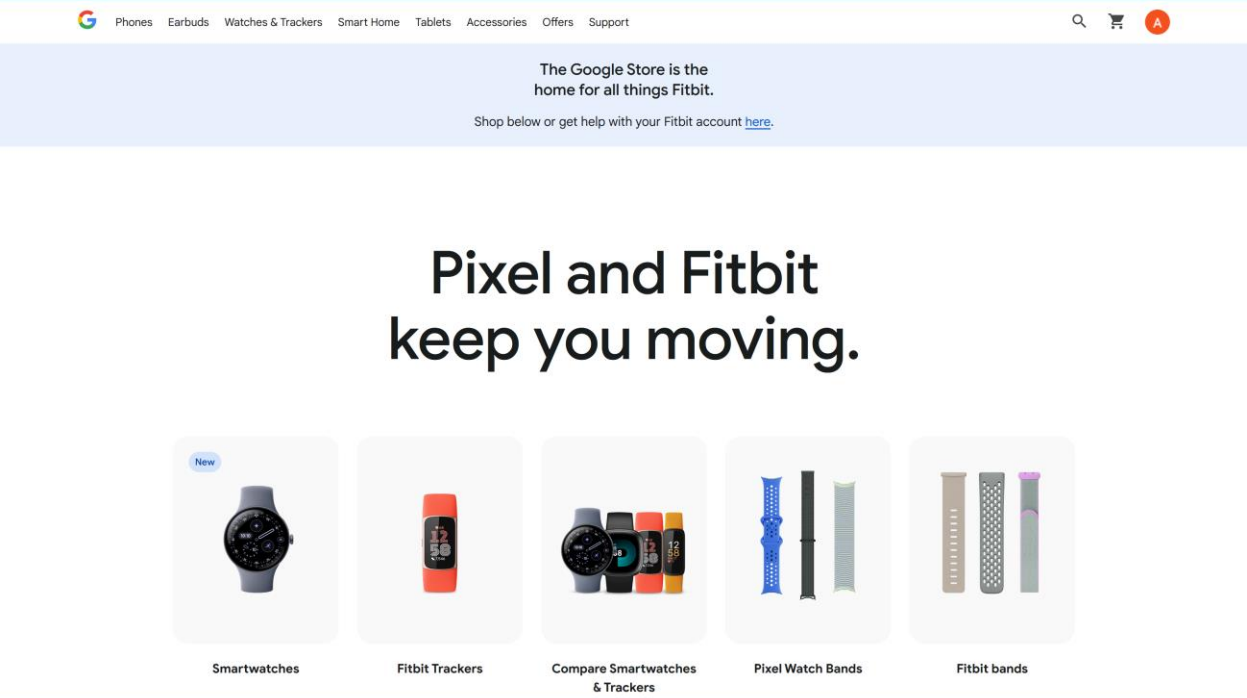


Figure 2: Fitbit Health

Strengths:

FitBit can provide real-time data collection to create an overview of health, provides a strong visualization of the data along fit its focus on integration with wearable devices such as watch, phone, wrist bands etc.

Limitations:

Although it provides such great value, it requires a wearable hardware device to make the most out of it. It also has limitations to diet planning along with all the good features locked behind a paywall which a normal user might not be able to afford.

2.3.3 Project 3: Ada Health

Ada Health is an AI-based symptom assessment tool designed to provide health insights based on user-reported symptoms.

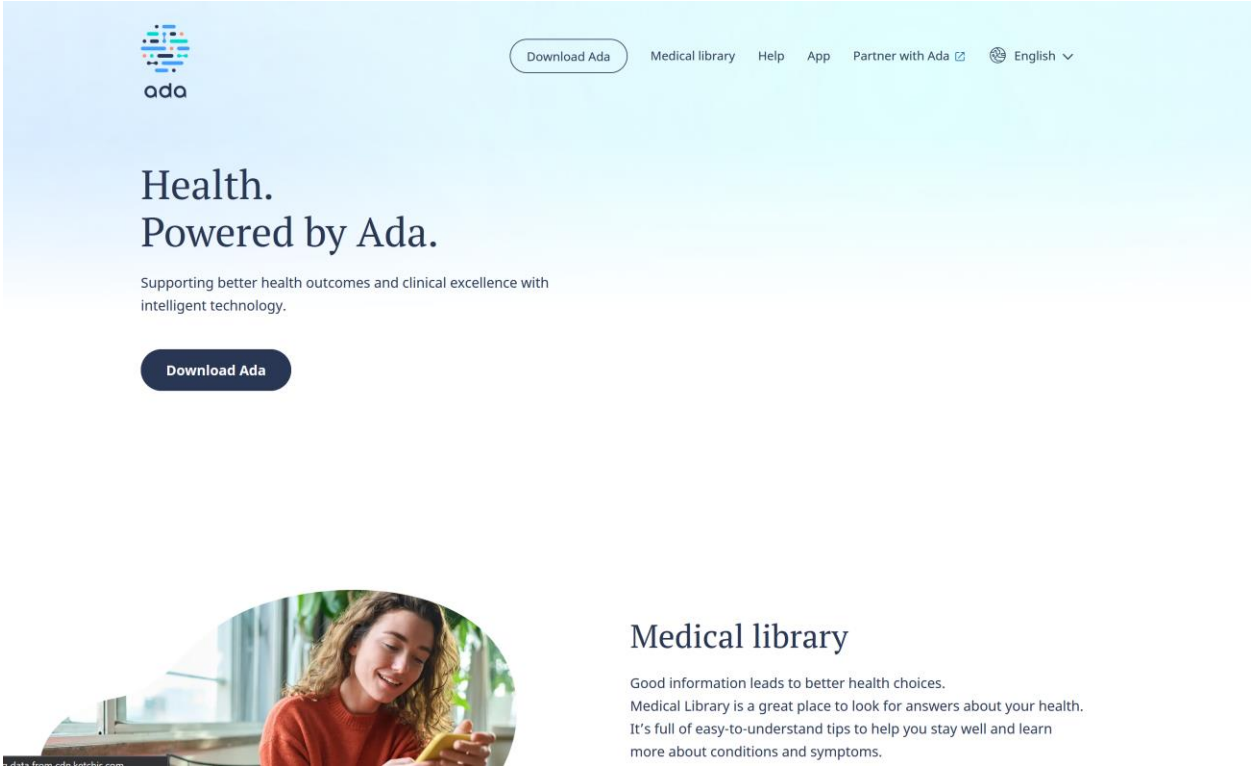


Figure 3: Ada Health

Strengths:

Ada health is more similar in terms of AI-driven health assessment. It provides a clear explanation for possible diseases. It is based on medical data, so most of the time it is very accurate.

Limitations:

Some of the drawbacks on Ada Health is that it mainly only focuses on symptom diagnosis. It doesn't provide any fitness or diet planning to the user, nor does it provide lifestyle-oriented recommendations.

2.4 Comparison Table of Similar Projects

Feature	My fitness Pal	Fitbit	Ada health	Dr. Jigree
User Health Profile	Yes	Yes	Yes	Yes
Ai based health scoring	No	Limited	Yes	Yes
Diet recommendation	No	No	No	Yes
Workout planning	No	Yes	No	Yes
Free access	Partial	Partial	Yes	Yes

2.5 Comparison Analysis of Similar Projects

While existing platforms provide valuable services, they focus on isolated aspects of health or restrict functionality through subscriptions. The proposed system differentiates itself by offering an integrated health portfolio combining diet, exercise, and health scoring within a free and open-source framework. This will position the project as a lightweight web application, yet a comprehensive solution for general wellness tracking and management.

3. Development Methodology

3.1 Considered Methodologies

1. **Waterfall Model**

   In the waterfall technique, all requirements are collected from customers and stakeholders at the start of the project. This initial phase sets the foundation for the entire project. Then, a sequential project plan is created that considers those requirements. Each project phase flows into the next, following steadily downward like a waterfall, thus the name waterfall model (Lucidchart, 2024).

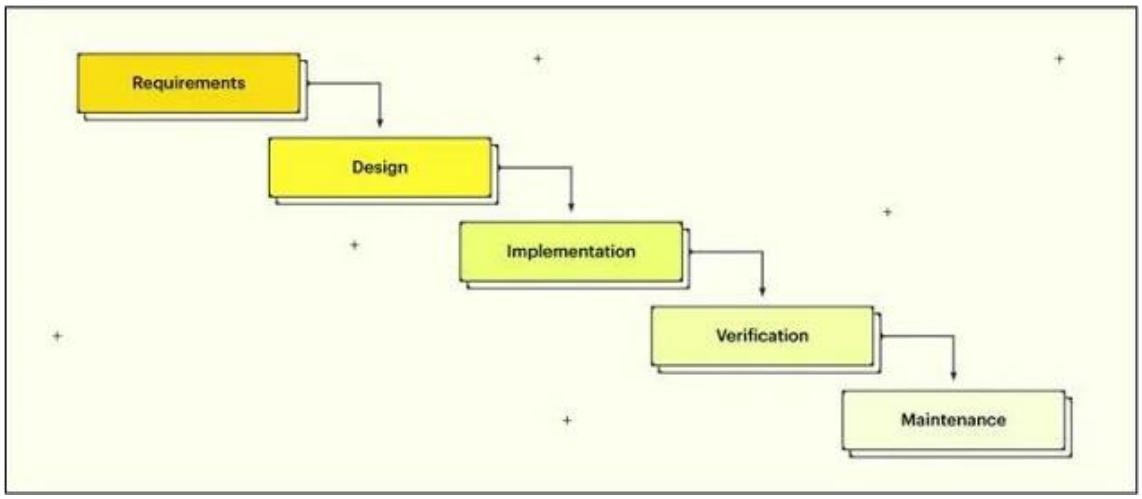


   Figure 4: Waterfall

2. **Scrum Model**

   Generally, Scrum is a simple framework that is a type of agile methodology for effective team collaboration on complex projects. It is a way for teams to work together to develop a product. Personal scrum is an agile methodology that adapts and applies scrum practices to one-person projects. This methodology is based on continuous learning and adaptation to changing circumstances. Periodic review, feedback, and adaptation are encouraged throughout the development process, with short release cycles enabling progressive growth (Scrum.org, 2024). Each sprint is equal and consists of 7 or 14 days each.

3. **Agile Methodology (Selected Methodology)** – Flexible, iterative, and well-suited for individual development. Iterative agile is best suited for individual projects like this. Agile methodology is a project management framework that breaks projects down into several dynamic phases, commonly known as sprints.

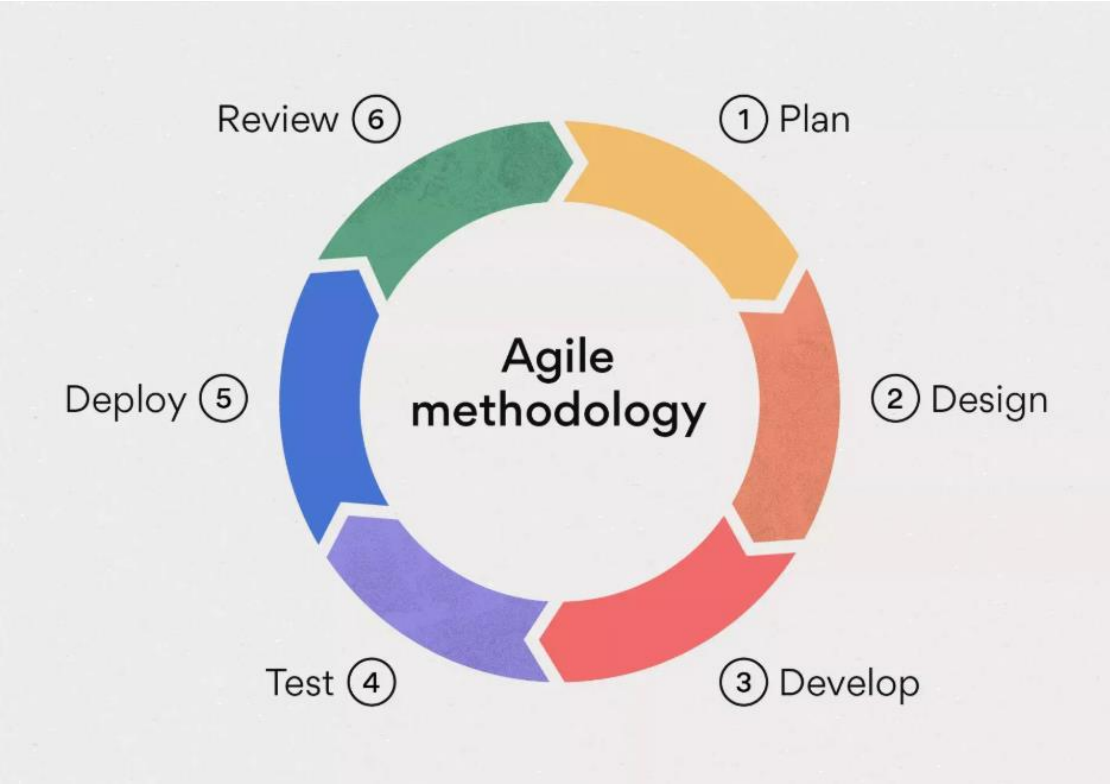


Figure 5: Agile

3.2 Selected Methodology

An Agile-inspired iterative methodology was selected. Development is divided into small increments, allowing continuous testing, feedback, and refinement of features. The development is done using individual agile methodology to cater to any changes that are needed as per the progress of the project.

3.3 Work Breakdown Structure

The work breakdown structure is tabled below.

Task	Duration	Description
Requirement Analysis and Dataset Collection	2 weeks	Gather requirements and download data to train the module. Preprocess and clean datasets
System Design	2 weeks	Architecture, UI mockups, ML pipeline
Backend Development	3 weeks	User authentication, profile store, etc.
ML Model Development	3 weeks	Training & integrating models
Frontend UI Development	2 weeks	UI pages such as dashboards, etc.
Integration Phase	1 weeks	Connection between frontend and backend.
Testing	2 weeks	Unit, functional, usability tests
Documentation & cleanups	1 week	User guide, system documentation, etc.

3.4 Gantt Chart

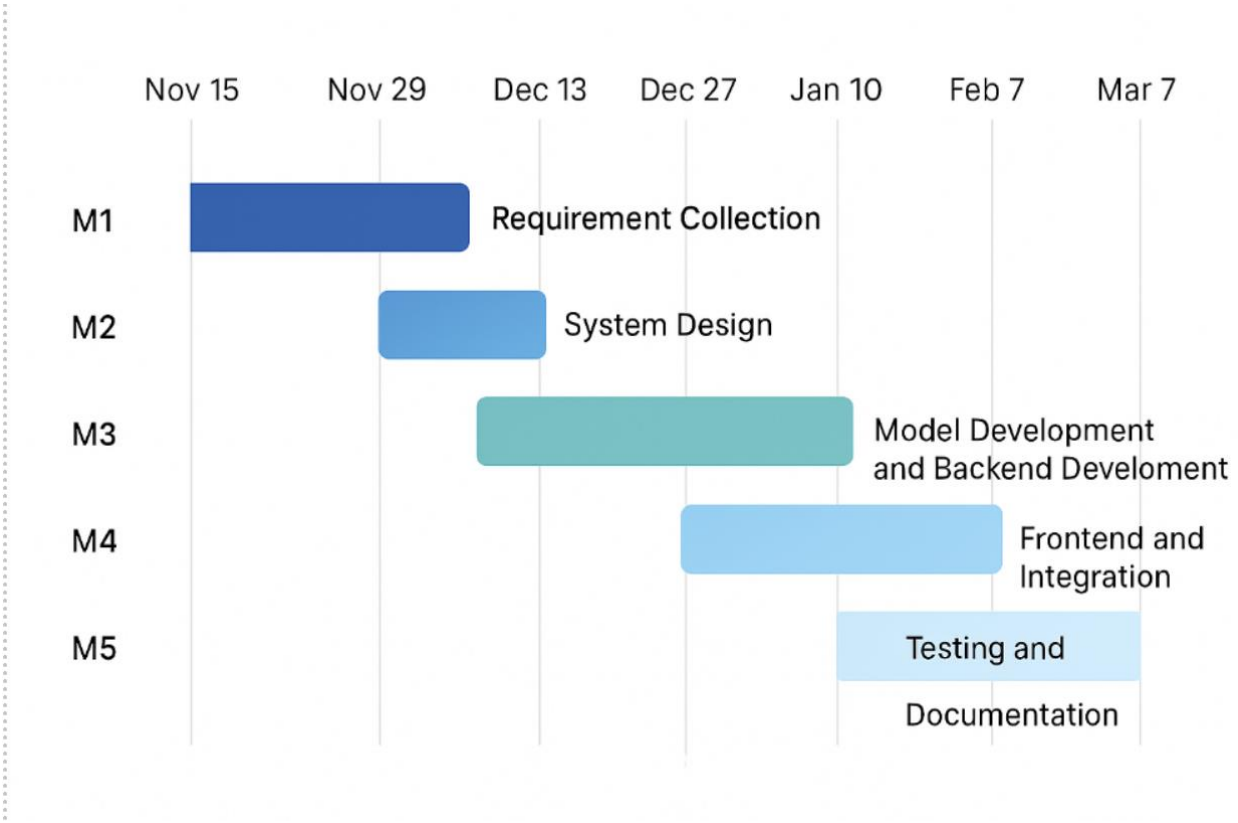


Figure 6: Gannt chart on timeline

4. Development Till Date

4.1 System Architecture

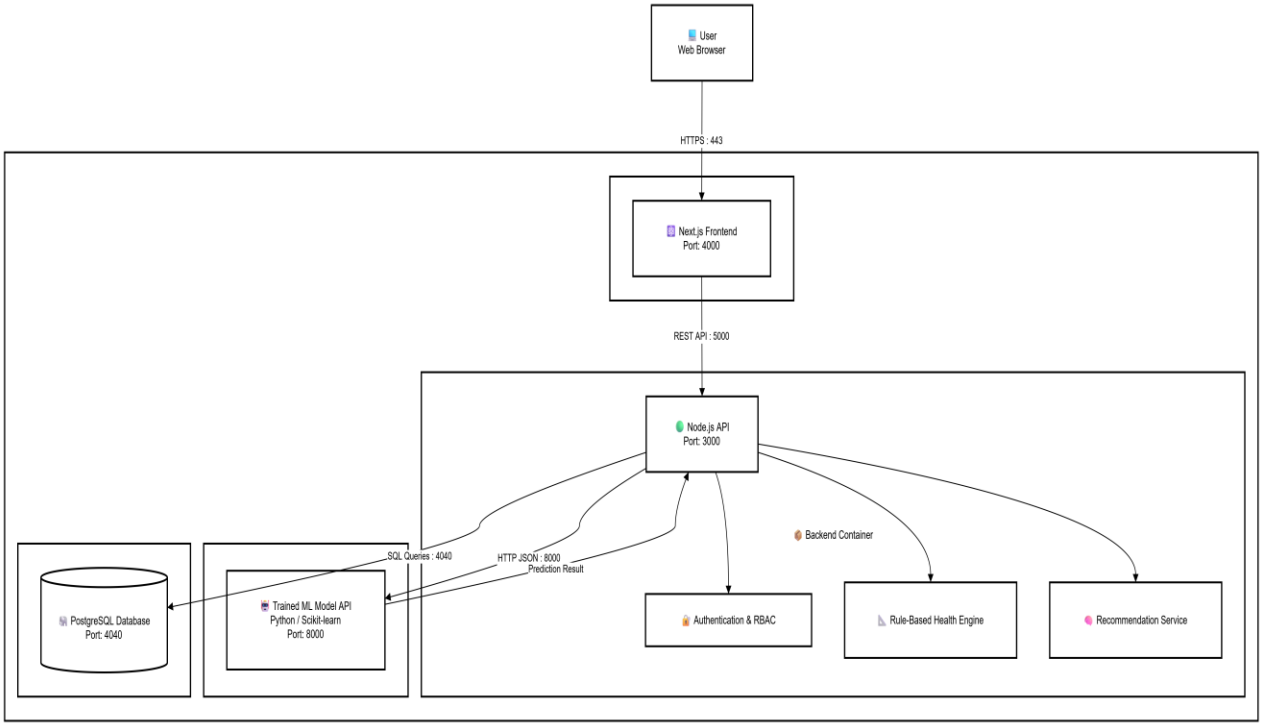


Figure 7: System Architecture design

There will be 4 components to the system that are very vital for the system.

1. **Frontend**: User will be communicating from the provided user interface of the system where they can interact seamlessly. There will be several intuative pages where anyone can sign up or login to enter their personal details. The profile screen will handle updating the user's info at any given time. They can then assess their health risk assessment. Based on the risk assessment and selected goal, they would get a personalized fitness plan as well as their dietary plan. Admin Screen would facilitate the user management system to the admins to monitor and maintain the users of the system. Further screens will be added in the future beyond MVP to expand the feature. They would also have access to a download as a pdf button where they can export the summary of the plan on demand. This would be hosted in the Port 4000 as a standard for the frontend in the industry.

2. **Backend**: All the logic behind the scenes would be handled by the backend. Each user will be assigned a temporary session upon successful login to ensure the security of the end-user. All the vital personal data would be encrypted before the storage to ensure security. Backend would be hosted on port 3000. Backend would interact with the ML module to feed and facilate the output to the user to ensure full-proof inputs. Data sanitizing is also vital to ensure no bad data is entered or stored in the database.
3. **ML**: Pretrained ML module is deployed in the port 8000 which would take the user data to process it to output the risk score as well as output the recommendation which then is stored in the database and presented to the user.
4. **Database**: All the data is stored in the NoSQL database where all the vital data is encrypted by backend. The encryption key is kept securely in .env variable only to be accessible by the system admin. The database would be hosted on port 4040 as shown in the diagram.

Docker

Docker is a set of products that uses operating-system-level virtualization to deliver software in packages called containers. Docker automates the deployment of applications within lightweight containers, enabling them to run consistently across different computing environments. Docker compose will be responsible for setting up all the tools of the project providing us an ideal container for the deployment. Initially, the project will be hosted in the docker before the deployment to ensure there is no difference in the environment.

4.2 UML Diagrams

4.2.1 AR Diagram (Application Flow)

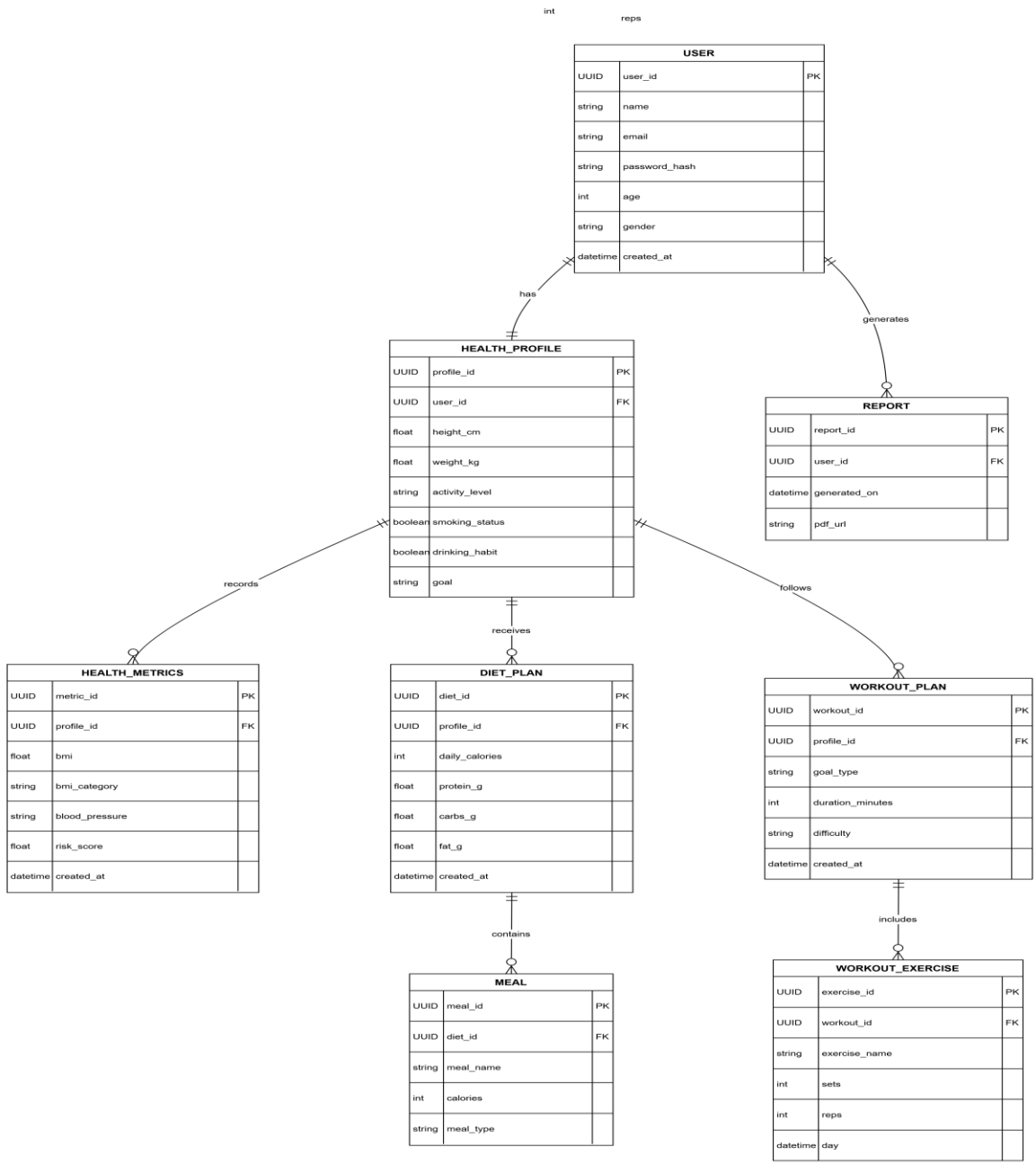


Figure 8: AR diagram on application flow

4.2.2 User Case diagrams

The use case diagrams for both the users and the Admin is outlined below.

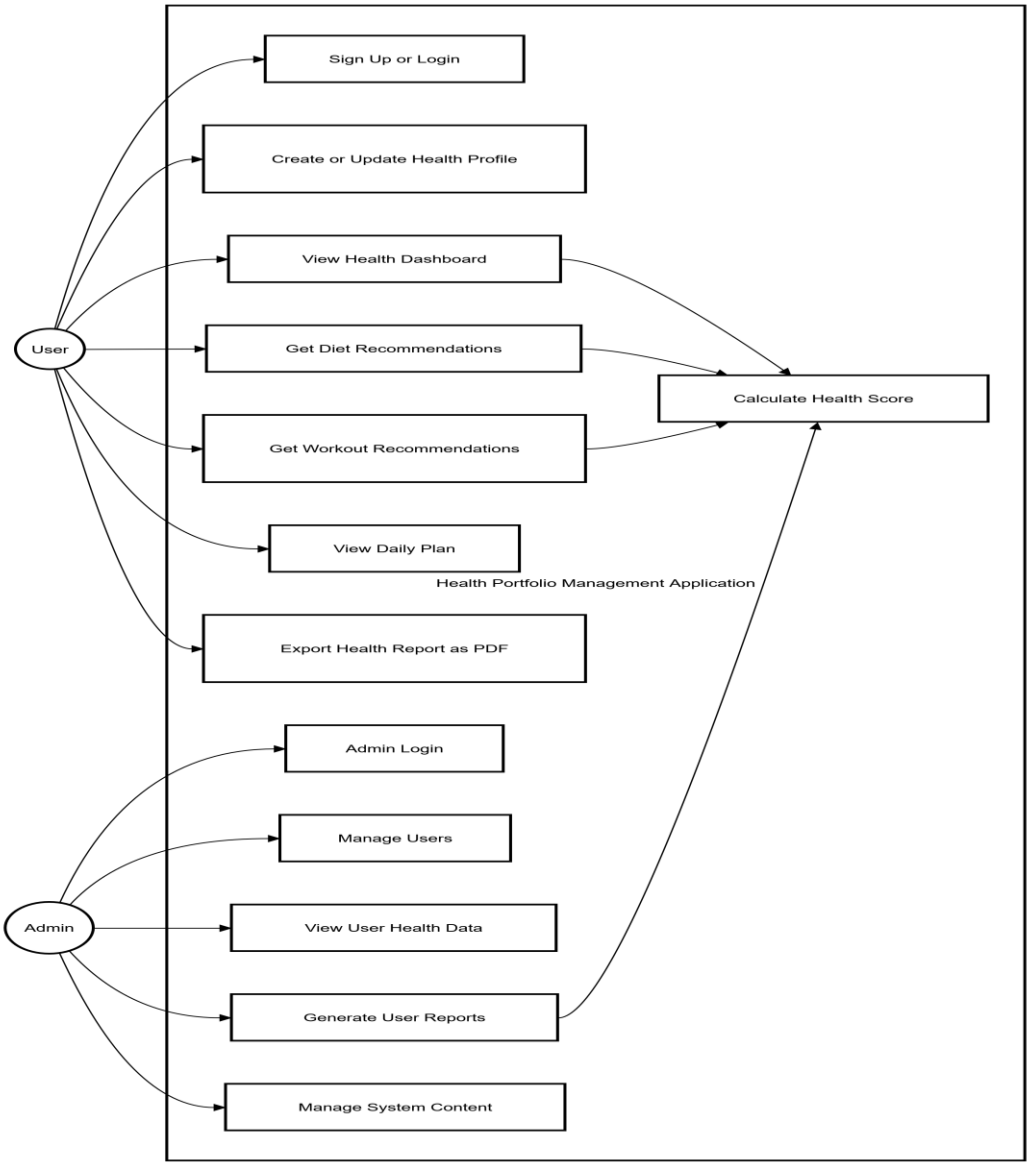


Figure 9: Use case diagram for User and Admin

4.2.3 Sequence and communication diagram

a. User registration

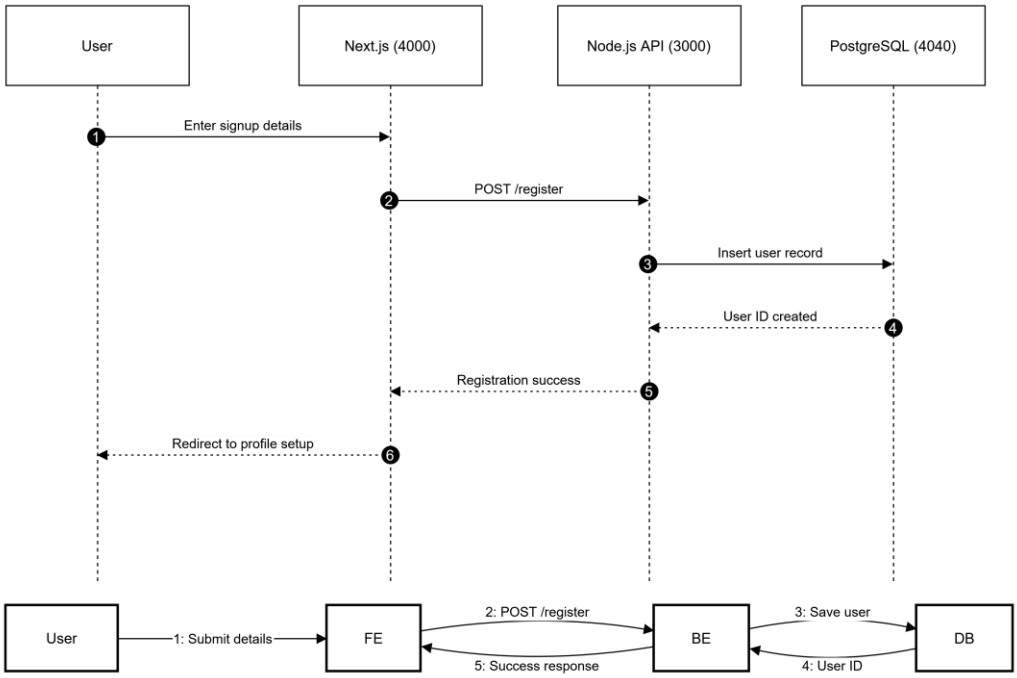


Figure 10: Sequence diagram for User registration

b. Profile update

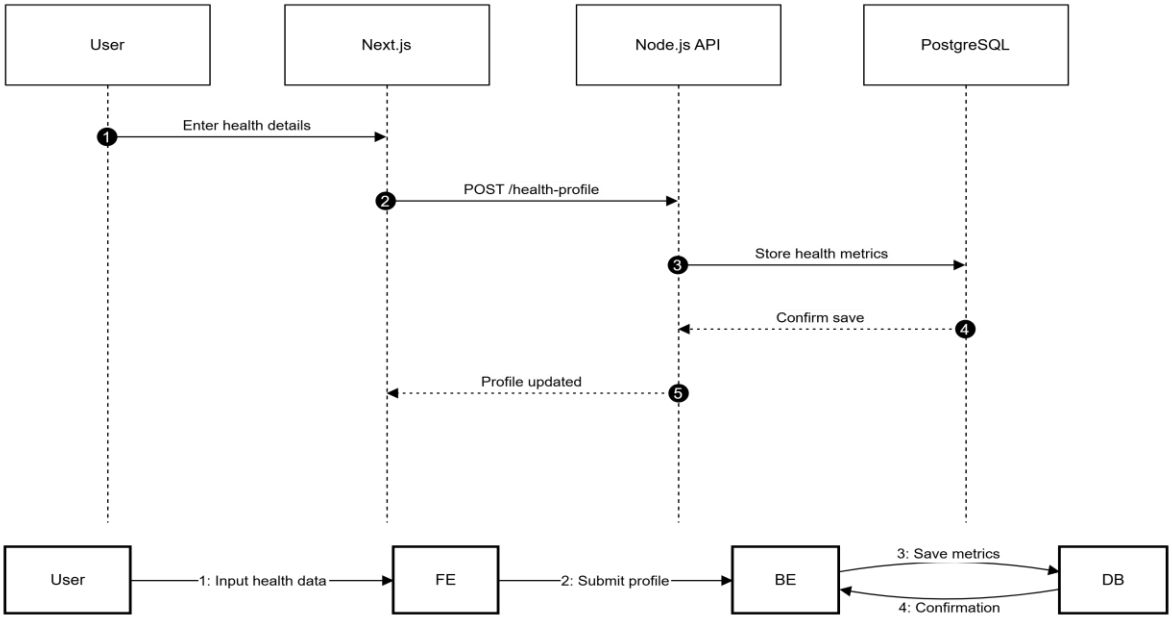


Figure 11: Sequence diagram for Profile update

c. Health Scoring

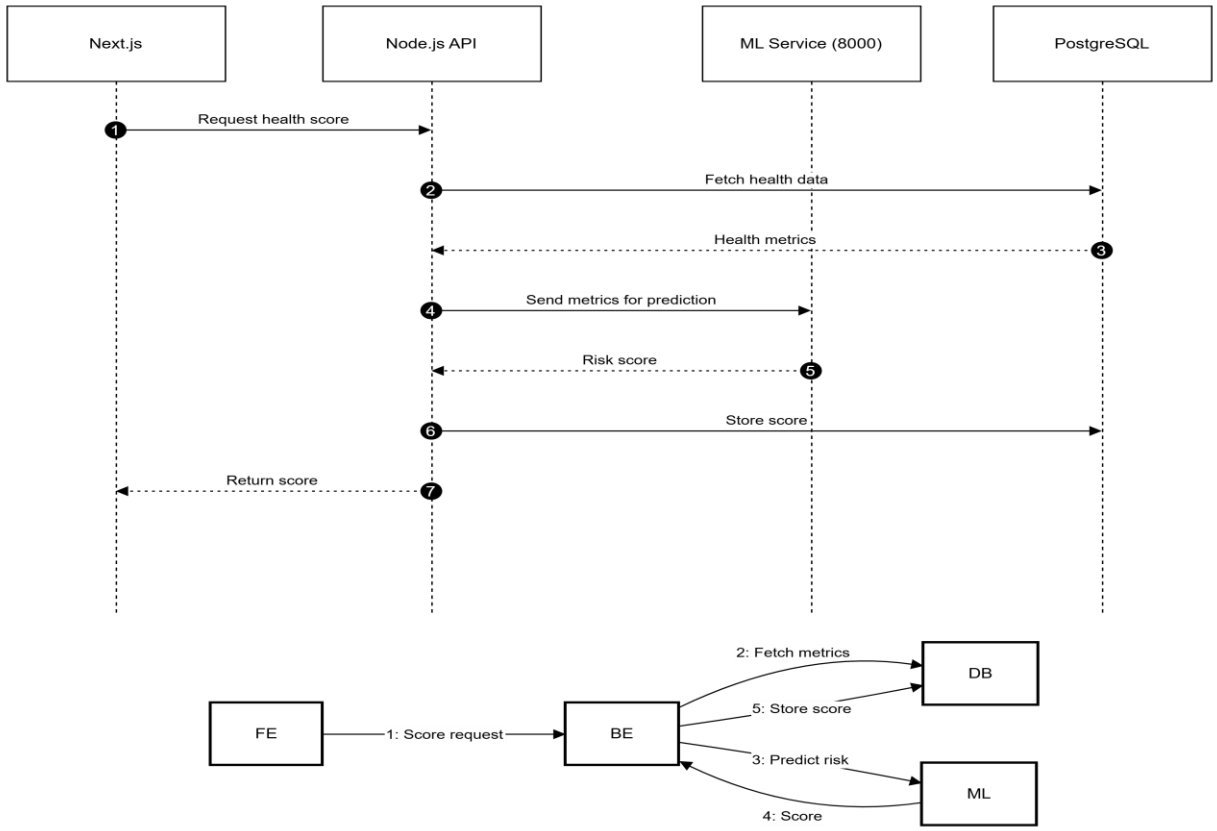


Figure 12: Sequence diagram for Health Scoring

d. Workout plan generator

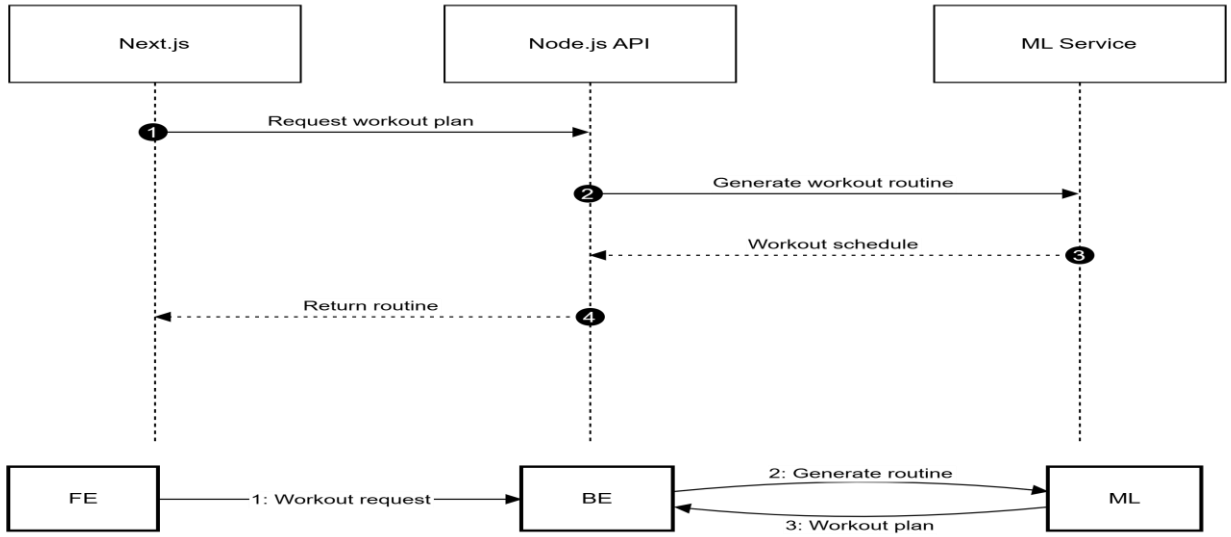


Figure 13: Sequence diagram for Workout plan generator.

e. Generate report

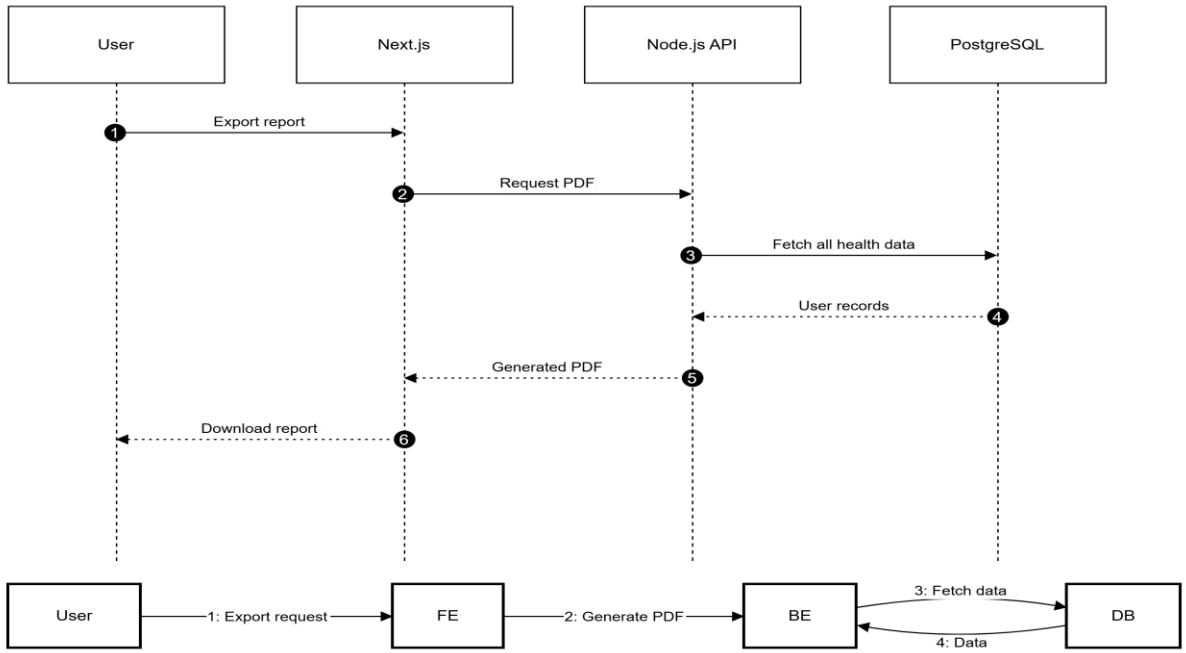


Figure 14: Sequence diagram for generating report

f. Admin user management

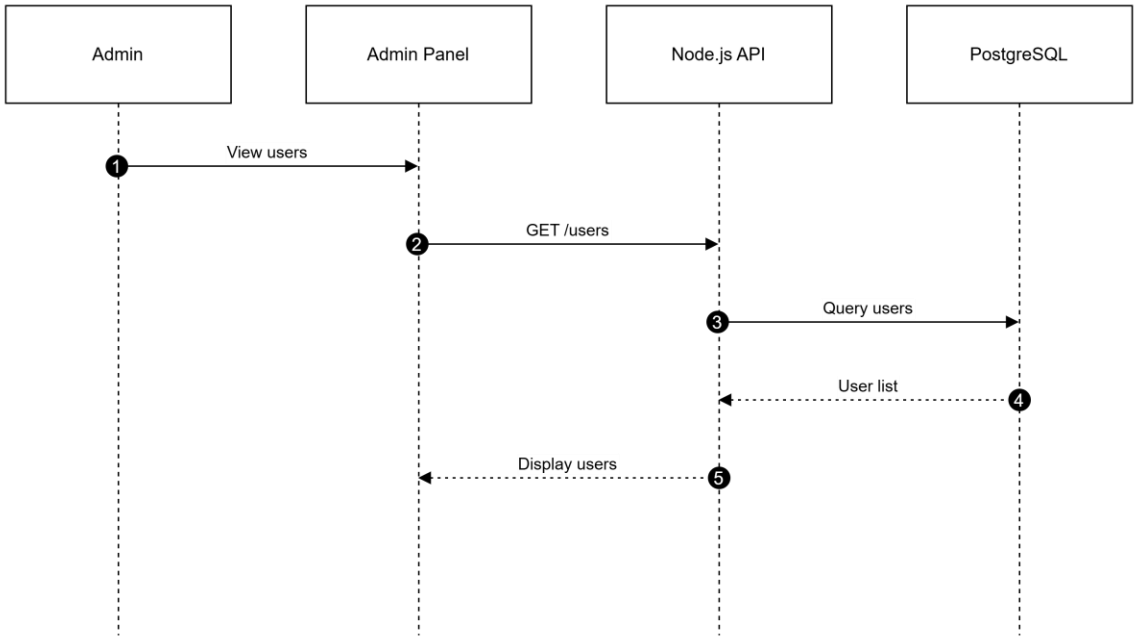


Figure 15: Sequence diagram for Admin user management

4.3 User Stories

- *As a user, I want to register to Dr. Jigree using my email address.*
- *As a user, I would like to enter and update my user profile details.*
- *As a user, I want to enter my health details so that I can understand my health status.*
- *As a user, I want personalized meal plans so that I can improve my diet.*
- *As a user, I want workout recommendations based on my goals.*
- *As a user, I want to export my health report as a PDF.*

4.4 Development Status with Evidence

a. Project repository created on GitHub

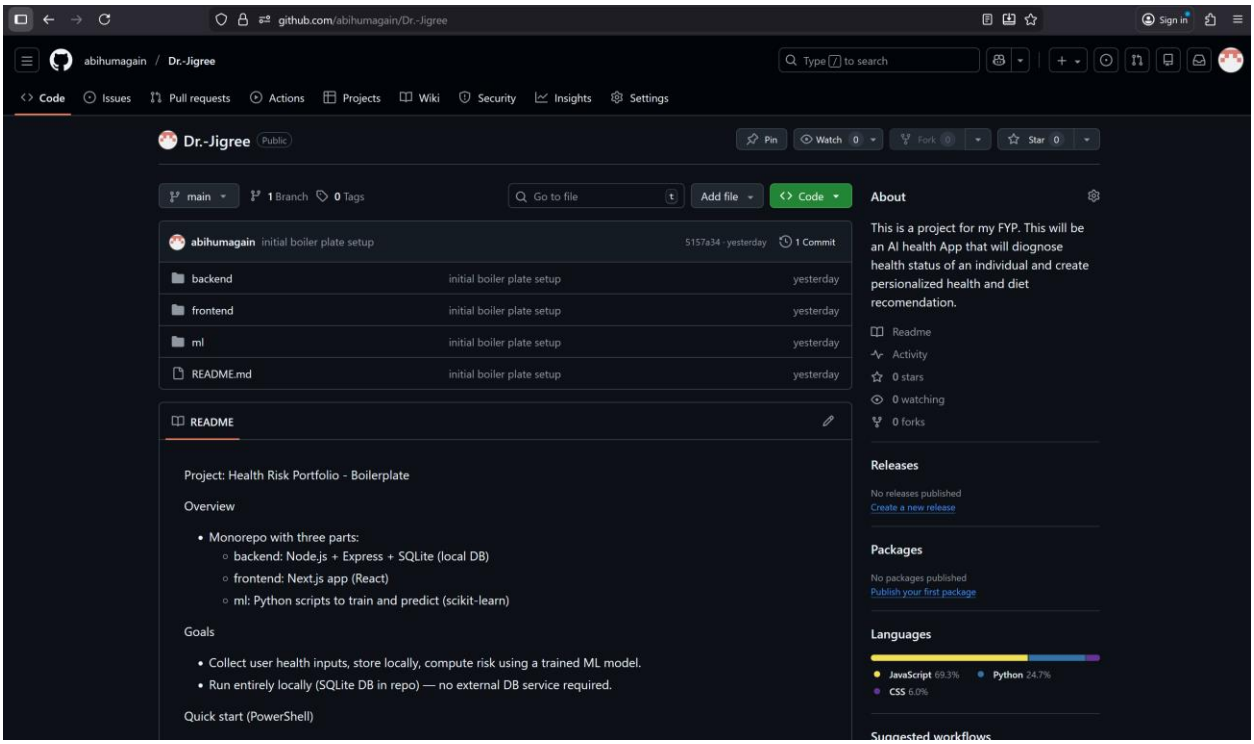
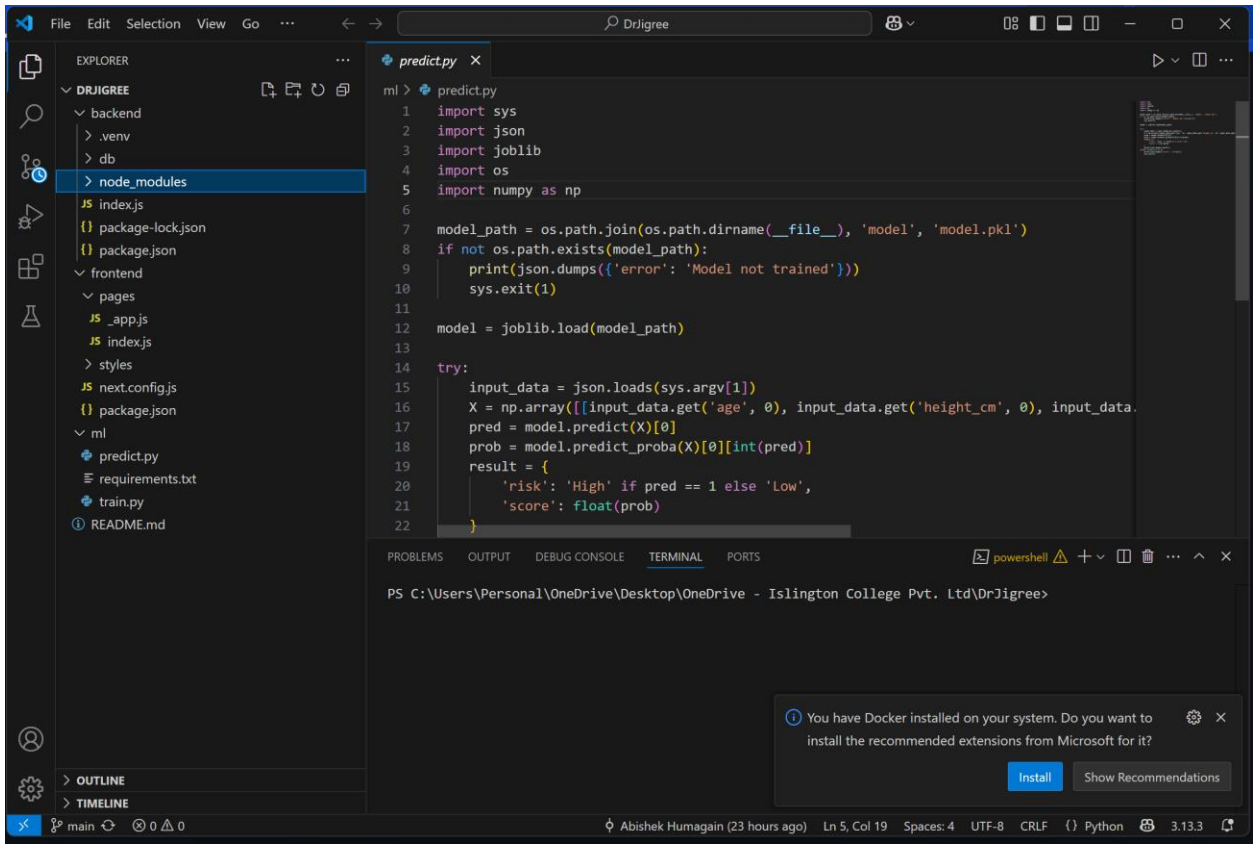


Figure 16: Github repo setup

b. Boiler plate setup



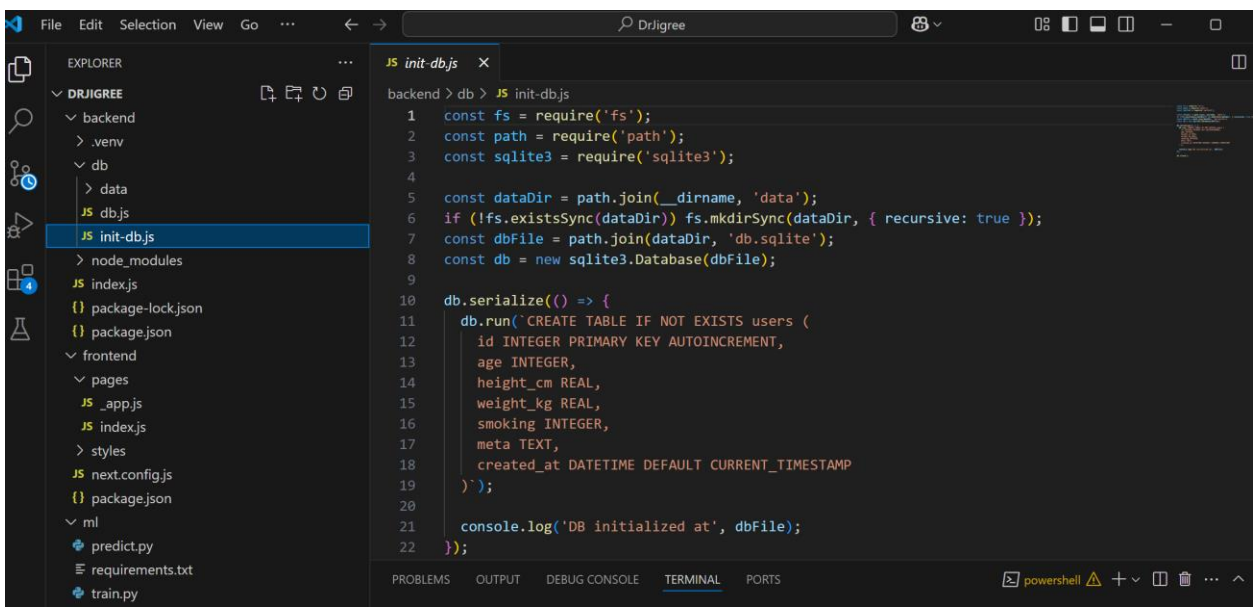


Figure 17: Bolier plate setup

c. Project and task setup in github

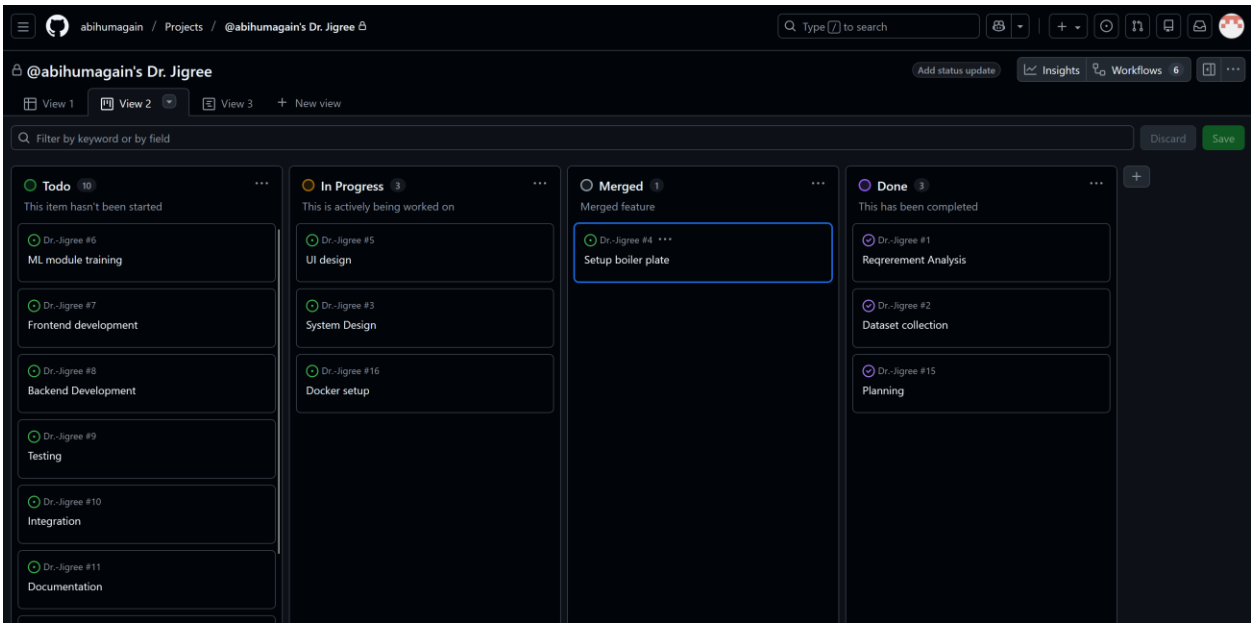


Figure 18: Project and task setup in Github

5. Analysis of Progress

5.1 Progress Review

The project is currently progressing as planned and even beyond the plans. Core architecture and foundational components are under development. The start is very strong with a steady and fast pace in development. Initially, there was a minor issue designing the system, but with further research the tools are finalized. If there is any issue or obstacle that were to arise, iterative agile would help to plan and deliver as per the changes.

5.2 Progress Table

S.N.	Tasks	Status	Progress
1	Topic selection	Completed	100%
2	Research on the topic	Completed	100%
3	Task breakdown	Completed	100%
4	System Design	Completed	100%

5	Set boiler plate	Completed	100%
6	UI design	Ongoing	30%
7	Backend development	Ongoing	10%
8	ML training	Not started	0%
9	Integration of FE, BE and ML	Not started	0%
10	Docker setup	Ongoing	50%
11	Support	Not Started	0%
12	Optimization	Not started	0%
13	Documentation	Not started	0%
14	Testing	Not started	0%
15	User Guide	Not started	0%
16	Cleanup	Not started	0%
17	Deployment	Not started	0%

5.3 Action Plan

The next steps are straight forward as per the backlog board shown above. Some of the upcoming tasks are listed below:

- Finish up the UI design
- Get the frontend setup to create the screens to provide intuitive user interface
- Backend development
- Database setup along with encryption at rest mechanism
- Complete ML model training and evaluation
- Integrate recommendation modules
- Begin system testing

6. Future Work

Currently the plan is to just deliver what was initially proposed as some of the features will be time consuming. Beyond MVP, future enhancements can include wearable integration, mobile application support, advanced predictive analytics, and multilingual support. Upon research, these features are time and effort consuming to deliver within the project timeline but are nice to have if I can get to it.

7. Conclusion

Dr. Jigree aims to address a growing global need for accessible and personalized health assistance. By using the machine learning-driven recommendations, the Health Portfolio Management Application will offer an affordable, practical, and user-friendly solution to help individuals improve their lifestyle and health. The planned methodology, technical approach, and structured timeline ensure that the proposed MVP can be delivered effectively within the academic year, with opportunities for future enhancement and real-world application.

This interim report demonstrates steady progress toward delivering a functional health portfolio management system. Background research and system analysis confirm the relevance of the project, and initial development milestones have been successfully achieved.

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